

# Jiyu Hu

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## Education

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| <b>University of Illinois Urbana-Champaign</b><br><i>Ph.D. in Computer Science</i>                            | Aug. 2023 – Present<br>Urbana, IL      |
| <b>Carnegie Mellon University – School of Computer Science</b><br><i>Master of Computational Data Science</i> | Aug. 2021 – May 2023<br>Pittsburgh, PA |
| <b>University of Illinois Urbana-Champaign</b><br><i>Bachelor of Science in Computer Engineering</i>          | Aug. 2017 – May 2021<br>Urbana, IL     |

## Publications

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\* = Equal Contribution

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| <b>SOSP '26</b>       | <b>Disk-Based LSMs: An Unexpectedly Good Index for Partly Coherent CXL Memory</b><br>Kiran Hombal, <a href="#">Jiyu Hu</a> , Marcos K. Aguilera, Aishwarya Ganesan, and Ramnatthan Alagappan<br><i>ACM SIGOPS 32th Symposium on Operating Systems Principles, 2026</i>                                                 |
| <b>OSDI '26</b>       | <b>MEGALON: Efficient Data Sharing for Partly Coherent CXL Memory</b><br><a href="#">Jiyu Hu</a> , Seokjoo Cho*, Landon Johnson*, Kiran Hombal, Shreesha G. Bhat, Marcos K. Aguilera, Ramnatthan Alagappan, and Aishwarya Ganesan<br><i>20th USENIX Symposium on Operating Systems Design and Implementation, 2026</i> |
| <b>TOCS</b>           | <b>LazyLog: A New Shared Log Abstraction and Design for Modern Low-Latency Applications</b><br>Xuhao Luo, Shreesha G. Bhat*, <a href="#">Jiyu Hu</a> *, Ramnatthan Alagappan, and Aishwarya Ganesan<br><i>ACM Transactions on Computer Systems, 2025</i>                                                               |
| <b>OSDI '25</b>       | <b>Low End-to-End Latency atop a Speculative Shared Log with Fix-Ante Ordering</b><br>Shreesha G. Bhat, Tony Hong, Xuhao Luo, <a href="#">Jiyu Hu</a> , Aishwarya Ganesan, and Ramnatthan Alagappan<br><i>19th USENIX Symposium on Operating Systems Design and Implementation, 2025</i>                               |
| <b>SOSP '24</b>       | <b>LazyLog: A New Shared Log Abstraction for Low-Latency Applications</b><br>Xuhao Luo, Shreesha G. Bhat*, <a href="#">Jiyu Hu</a> *, Ramnatthan Alagappan, and Aishwarya Ganesan<br><i>ACM SIGOPS 30th Symposium on Operating Systems Principles, 2024</i>                                                            |
| <b>HotStorage '24</b> | <b>Rethinking Erasure-Coding Libraries in the Age of Optimized Machine Learning</b><br><a href="#">Jiyu Hu</a> , Jack Kosaian, and K. V. Rashmi<br><i>16th ACM Workshop on Hot Topics in Storage and File Systems, 2024</i>                                                                                            |
| <b>TNSM</b>           | <b>Medley: A Membership Service for IoT Networks</b><br>Rui Yang, Jiangran Wang, <a href="#">Jiyu Hu</a> , Shichu Zhu, Yifei Li, and Indranil Gupta<br><i>IEEE Transactions on Network and Service Management, 2022</i>                                                                                                |

**SEC**                      **B2-Coupon: Efficient and Non-intrusive Mobile Coupon Distribution using Dual Bloom Filter**  
Xueda Shen\*, Jiyu Hu\*, Yunqi Zhang, and Ian C. Quinn  
*IEEE/ACM Symposium on Edge Computing, 2020*

## Presentations

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**Rethinking Erasure Coding Libraries in the Age of Optimized Machine Learning**                      Jul. 2024  
*HotStorage '24*                      Santa Clara, CA

**B<sup>2</sup> Coupon**                      Nov. 2020  
*ACM SEC (workshop talk)*

## Awards & Scholarships

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**Jay Lepreau Best Paper Award Nominee (MEGALON)**  
*OSDI '26*                      2026

**Student Travel Grant**  
*OSDI '25 & OSDI '26*                      2025, 2026

**Best Paper Award (LazyLog)**  
*SOSP '24*                      2024

**Senior Design Instructor's Award (The Best Senior Design Award)**  
*University of Illinois Urbana-Champaign*                      2021

**Bradley A. Simons Memorial Scholarship**  
*University of Illinois Urbana-Champaign*                      2019

**Dean's List**  
*University of Illinois Urbana-Champaign*                      2017, 2018, 2019, 2020, 2021

## Research Experience

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**Prism**                      Jun. 2025 – Jun. 2026  
*DASSL, University of Illinois Urbana-Champaign*                      Urbana, IL

- Investigate efficient index design on CXL memory under the assumption of partial hardware coherence.

**MEGALON**                      Apr. 2024 – Dec. 2025  
*DASSL, University of Illinois Urbana-Champaign*                      Urbana, IL

- Investigate the design of efficient sharing of CXL memory under the assumption of partial hardware coherence.

**SpecLog**                      Apr. 2024 – Dec. 2024  
*DASSL, University of Illinois Urbana-Champaign*                      Urbana, IL

- Design and implement a new shared log abstraction that significantly decreases e2e latency by speculation.

**LazyLog**                      Jan. 2024 – Jul. 2024  
*DASSL, University of Illinois Urbana-Champaign*                      Urbana, IL

- Design and implement a new shared log abstraction that significantly decreases append latency from state-of-the-art shared log implementations by delaying ordering the appended log entries.

**TVM-EC**                      Jan. 2022 – Jul. 2024  
*TheSys Lab, Carnegie Mellon University*                      Pittsburgh, PA

- Propose a new way of implementing high-performance erasure-coding libraries via machine learning libraries, reducing the effort to design and maintain erasure coding libraries.

**Medley**                      Jan. 2020 – Jul. 2021  
*Distributed Protocols Research Group, University of Illinois Urbana-Champaign*                      Urbana, IL

- Develop and evaluate a new IoT failure detection protocol that is aware of the spatial locality of physical nodes so as to decrease the overall communication overhead in an unstable network environment.

**B<sup>2</sup>-Coupon**

*Prof. Dong Xuan's Research Group, The Ohio State University*

Nov. 2019 – Nov. 2020

*Columbus, OH*

- Design a better coupon distribution protocol for mobile devices.

## Services

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**CAIS '26**, Artifact Evaluation Committee

**EuroSys '26**, Shadow PC

**SOSP '25**, Artifact Evaluation Committee

**FAST '26**, Artifact Evaluation Committee

**OSDI '24**, **ATC '24**, Artifact Evaluation Committee

## Teaching

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**CS 498 Cloud Computing Applications**

*University of Illinois Urbana-Champaign*

TA  
2026

## Industry Experience

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**GSL, Microsoft Corporation**

*Research Intern (Supervisor: Rathijit Sen)*

May 2026 – Aug. 2026

- Work on disaggregated memory for database systems.

**USM, Exadata, Oracle America, Inc.**

*Software Intern*

May 2022 – Aug. 2022

- Worked on the kernel log aggregation and analysis framework of Oracle distributed database.

**CUDA Math Library, NVIDIA Corporation**

*Software Intern*

June 2020 – Aug. 2020

- Analyzed the floating point error propagation in cuBLAS GEMM kernel.